

Spine Issues in Skeletally Immature Athletes

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There are many challenges facing the sports physician in the care of the spine in pediatric athletes. They may be asked to evaluate a child or adolescent with a known condition and assess whether he or she can safely participate in athletic activities. Second, when an acute traumatic event has occurred, sports physicians may be asked to evaluate and provide initial management for the injured athlete. In addition, sports physicians manage and treat conditions that create chronic pain or deformity and may have an impact on sports participation.

The anatomic differences between the mature and growing spine can make these varied roles more complicated. In addition to injuries that are observed in both children and adults, the growing spine leads to injuries that are unique to the pediatric and adolescent populations. This chapter describes the relevant anatomy and related injuries observed in the young athlete, as well as key components in the evaluation of these spine conditions and their management. In addition, guidelines for both restriction from sports and timing of return to athletic activities are provided.

ANATOMY

Fracture patterns and biomechanics, particularly in the cervical spine, differ for children based on age. In younger children, the fulcrum is at C2-C3 because of their proportionally large skulls. Consequently, for children younger than 8 years, 87% of cervical spine injuries occur at or above C3 and 50% are associated with head injuries.¹ Older children have fracture patterns similar to those of adults, with the subaxial spine most commonly affected.

Diagnosis of cervical spine injuries in the child can be challenging, particularly in the younger age groups, because much of the cervical spine has not yet ossified. The anterior ring of C1 does not ossify until after the age of 1 year, making detection of injury in this region especially challenging (Fig. 140.1). Between the ages of 3 and 6 years, fusion of the dens to the neural arches and anterior body occurs, which can be mistaken for a fracture at the base of the odontoid. By age 6 years, the spinal canal of the cervical spine has reached adult dimensions. Lateral radiographs show a progressively increasing facet angle from birth to 8 years of age that allows motion through flexion and extension and contributes to the appearance of pseudosubluxation often seen at C2-C3 and C3-C4. Pseudosubluxation can be differentiated from true subluxation on plain radiographs

by the maintenance of the Swischuk line, which is a straight line drawn along the spinolaminar line (the anterior edge of the posterior neural arch; Fig. 140.2). However, care should be taken as while pseudosubluxation is a normal finding and does not require treatment, injuries can also occur at this level (Figs. 140.3A and B). In addition, incomplete ossification often gives the appearance of wedging of the cervical vertebral bodies on lateral radiographs until age 10 years. The atlantodens interval (ADI) may be increased in children. As opposed to adults, in whom a normal ADI is less than 3 mm, in young children the ADI may be up to 4.5 mm without signifying injury.

Additional considerations also exist in the pediatric thoracolumbar spine compared with the adult spine. The ring apophysis that extends peripherally around the vertebral body is adherent to the annulus of the intervertebral disk and can be at risk for injury. These injuries may be mistaken both clinically and radiographically for a pediatric disk herniation.

GENERAL HISTORY AND PHYSICAL EXAMINATION FOR EVALUATION OF THE PEDIATRIC SPINE

The sports physician may be evaluating the spine of a pediatric patient in preparation for participation in athletic activities, after an acute event that occurs while participating in a sport, or when the patient has chronic pain. It is important to differentiate the more common generalized aches and pains that accompany both athletic training and adolescence from more serious causes of pain. Atypical pain may be indicative of a more serious injury, a tumor, infection, or another more severe clinical entity. Box 140.1 outlines key elements in the history that can guide the clinician in making this distinction.

In evaluating spinal injuries, the mechanism and impact can alert the clinician to more critical conditions. In acute scenarios in which a potentially unstable spine injury is suspected, proper immobilization of the spine is necessary to prevent worsening of the injury. For children younger than 8 years, a pediatric backboard with a recess should be used to accommodate their proportionally large head size and prevent additional injury from the acute flexion that can be caused by a traditional backboard.

The physical examination is critical but also challenging in the child with suspected spinal injury. In cases in which spinal instability is not suspected, the focused neurologic examination

Abstract

Skeletally immature athletes are uniquely vulnerable to a variety of spine conditions that the sports physician may be required to evaluate. An understanding of normal spinal anatomy through growth and development is critical for understanding these conditions and their management. This will enable a better understanding of the proper evaluation and treatment of these athletes both in the office or following an acute traumatic event.

We will review key history and physical exam findings to aid in appropriately diagnosing common spinal injuries that may impact these young patients, such as spondylolysis and apophyseal endplate fractures, and will also discuss their treatment. In addition, we will review spine conditions that may impact a pediatric patient's ability to safely participate in athletic activities, such as Klippel Feil syndrome or Down syndrome patients with atlantoaxial instability. In addition to describing the evaluation and treatment of both acute and chronic spinal conditions impacting the skeletally immature athlete, we will also outline the appropriate steps to safely optimize their return to sports participation.

Although there are unique challenges and concerns with the evaluation of the growing spine in an athlete, with an understanding of these distinct clinical conditions and their management, many of these young patients can enjoy continued sports participation.

Keywords

spondylolysis
apophyseal endplate fractures
Chance fractures
scoliosis
atlantoaxial instability
Klippel-Feil syndrome
rotatory atlantoaxial subluxation

outlined (Box 140.2) provides an efficient assessment and can be performed even in young children.

SPONDYLOLYSIS/SPONDYLOLISTHESIS

Spondylolysis is a defect in the pars interarticularis and is most commonly found at the L5 level. It is by far the most common

cause of back pain, accounting for up to 47% of cases in the adolescent population in some series.² The prevalence of spondylolysis is approximately 6% among the general population.³ Approximately 80% of cases are bilateral, whereas 20% of cases are unilateral. When a patient has an associated slippage of a vertebra in relation to the adjacent vertebra, the term “spondylolisthesis” is used.



Fig. 140.1 Detection of cervical spine injuries or abnormalities in the young child can be challenging, because much of the cervical spine has not yet ossified. The anterior ring of C1 does not ossify until after the age of 1 year and is not visible on plain radiographs at that age, as depicted here.

BOX 140.1 Pediatric Back Pain Red Flags

The following pain characteristics are not associated with typical muscular back pain and warrant further investigation.

- Night pain
- Pain in young children, especially if they stop playing
- Frequent pain at rest, not including sitting (pain with prolonged sitting in school is extremely common in teens with mechanical low back pain)
- Back pain associated with an abnormal gait, limp, unsteadiness on feet, and so on.
- With associated constitutional symptoms (e.g., fever, lethargy, weight loss, loss of appetite)
- Atypical location, especially thoracic rather than lumbar pain (although pain in the trapezius/periscapular muscles is often due to backpacks)
- History of worsening pain
- With neurologic symptoms or physical findings (cavus foot, etc.)
- Kyphosis
- With important cutaneous findings (e.g., high and deep dimple, hairy patch, café-au-lait spots suggestive of neurofibromatosis)
- Positive finger test (child localizes pain to area the size of a fingertip)
- Positive coin test (have child keep the back straight when picking something up)
- Listing to side on Adams forward bend
- Tight hamstrings (popliteal angle >50 degrees)

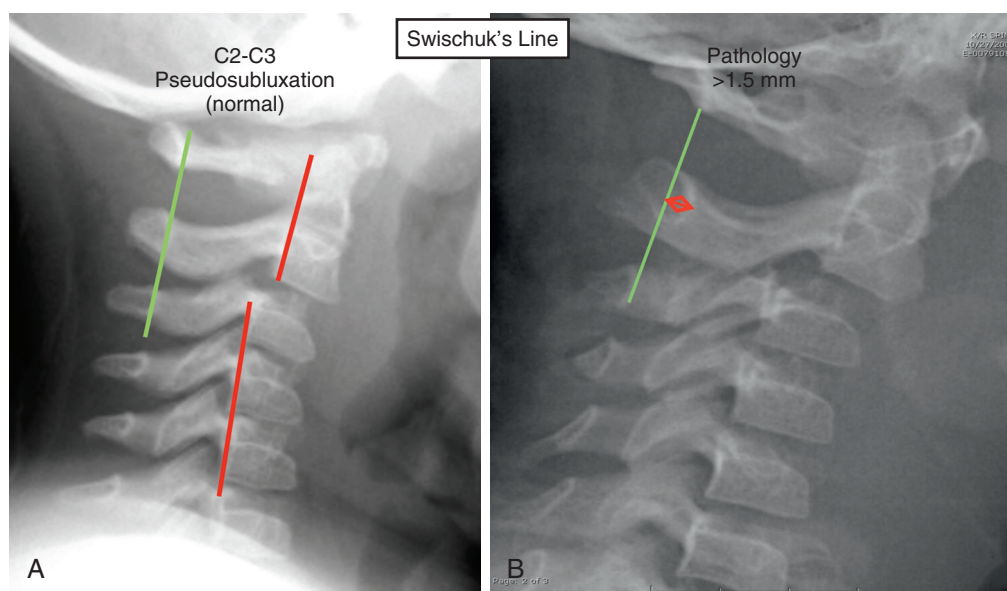


Fig. 140.2 Pseudo-subluxation can be differentiated from true subluxation on plain radiographs by the maintenance of the Swischuk line (green), which is a straight line drawn along the anterior edge of the posterior neural arch. (A) The patient has no acute injury, which is confirmed by the maintenance of this line. (B) The patient has a change in the relation of the posterior elements to this line, which represents true subluxation. (Courtesy Children's Orthopedic Center, Los Angeles, CA.)

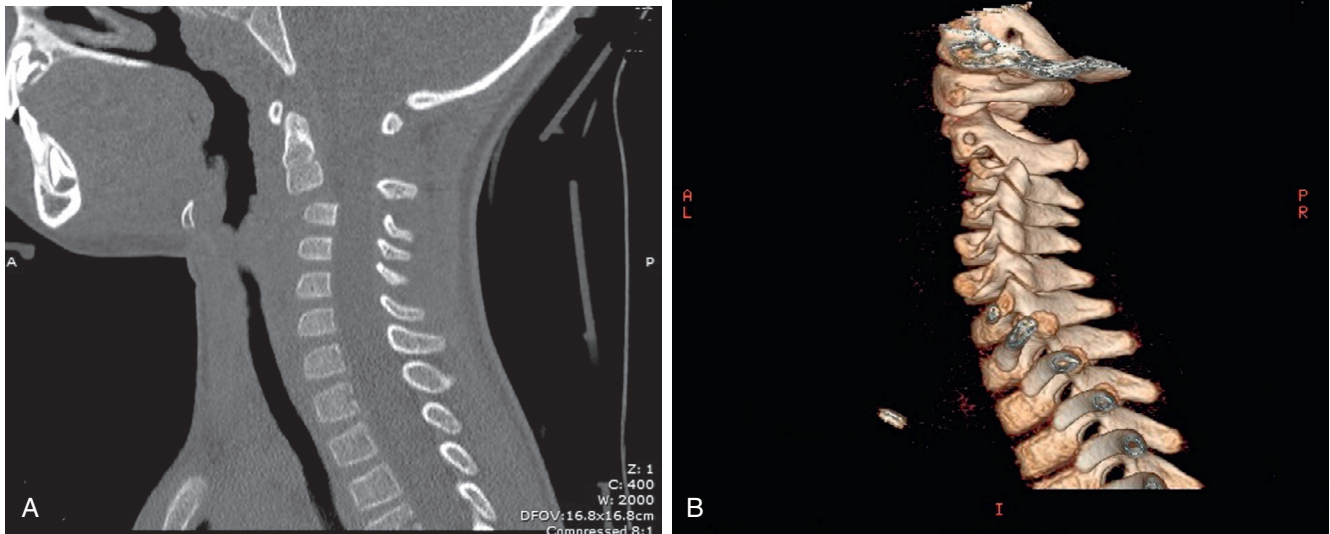


Fig. 140.3 (A and B) Although pseudosubluxation is a normal finding that does not require treatment, true injuries can also occur at this level. Sagittal and 3D computed tomography scan images of a 3-year-old involved in a high-speed motor vehicle collision demonstrate jumped facets at C2-C3. Magnetic resonance imaging confirmed extensive associated ligamentous injury.

BOX 140.2 60-Second Neurologic Examination

Hop on each foot, one at a time
Walk on heels
Reflexes
Foot inspection
Ankle dorsiflexion to assess muscle tone and clonus
Sensation
Popliteal angle

From Skaggs DL, Flynn JM. *Staying Out of Trouble in Pediatric Orthopaedics*. Philadelphia: Lippincott Williams & Wilkins; 2005.

Classification

Spondylolisthesis can be classified on the basis of either the type of the spondylolisthesis or the degree of displacement. The five types of spondylolisthesis are dysplastic, isthmic, degenerative, traumatic, and pathologic.^{4,5} Of these, dysplastic (in which facet joints allow anterior translation) and isthmic (associated with a lesion of the pars interarticularis) types are most commonly encountered in children. Spondylolisthesis can also be classified by the degree of displacement in relation to the adjacent vertebra with use of the Meyerding grading system (shown in Fig. 140.4, with an example of a grade 3 spondylolisthesis), where 1% to 24% = 1, 25% to 49% = 2, 50% to 74% = 3, and 75% to 100% = 4. Cases in which the amount of displacement exceeds 100% are referred to as *spondyloptosis*.

History and Physical Examination

Spondylolysis is associated with sports such as gymnastics, volleyball, football, diving, and pole vaulting, in which repetitive hyperextension occurs, but can be seen in essentially any sport. A typical history is one of activity-related back pain that is well localized to the lower spine, with radiating pain in the buttocks or legs in some cases. Approximately 40% of patients recall a traumatic event at the onset of pain.⁶



Fig. 140.4 An example of grade 3 spondylolisthesis using the Meyerding grading system. This system describes the percentage of translation of the superior vertebral body relative to the inferior vertebral body. Grade 1 = 1% to 24%; grade 2 = 25% to 49%; grade 3 = 50% to 74%; and grade 4 = 75% to 100%. Cases in which the amount of displacement exceeds 100% are referred to as spondyloptosis. (Courtesy Children's Orthopedic Center, Los Angeles, CA.)

On physical examination, patients may have lumbar hyperlordosis. Conversely, they may have flattening through the lumbar region in cases of severe pain or a high-grade spondylolisthesis.⁷ The classic physical exam finding is pain with standing spine hyperextension, especially during single-leg stance or with concomitant twisting. Hamstring contractures are commonly associated with spondylolysis/spondylolisthesis. In some severe cases,

this condition leads to the abnormal gait pattern described by Phalen and Dickson,⁸ consisting of crouching, a short stride, and an incomplete swing phase.

Imaging

A great deal of controversy exists regarding the optimal imaging modality for evaluation of spondylolysis. Although the characteristic “collar of the Scottie dog” observed on oblique radiographs is often difficult to appreciate, plain radiographs are commonly used as first-line imaging (Fig. 140.5). In some series, more than half of spondylolysis lesions (53%) can be missed on plain films alone.⁹ A single photon emission computed tomographic (CT) bone scan is more sensitive for detecting defects of the pars but

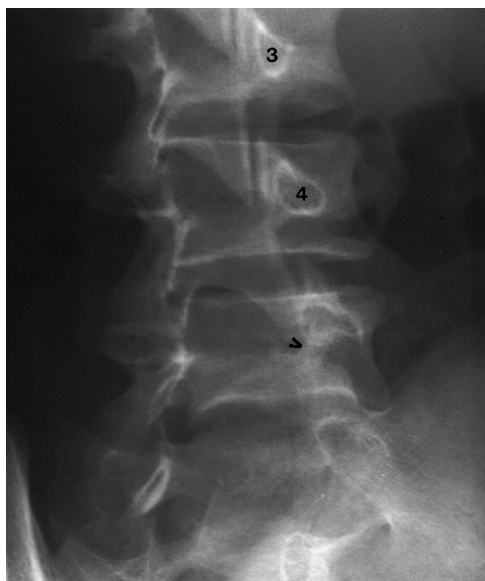


Fig. 140.5 The characteristic “collar of the Scottie dog” observed on oblique radiograph. (Courtesy Children’s Orthopedic Center, Los Angeles, CA.)

is nonspecific and thus is positive for other pathologic conditions, including infection, osteoid osteomas, and neoplasms.¹⁰ Single photon emission CT may also miss older, “cold” lesions. Consequently, our practice is to obtain a spot lateral of L5-S1 and then move directly to a limited CT if the signs and symptoms are consistent with a spondylolysis.

Since the advent and development of CT, many studies have verified that CT is more sensitive than plain radiographs in detecting early spondylolytic lesions.¹¹ In addition, even in patients with radiographs that demonstrate spondylolysis, a CT scan may be helpful for treatment planning to assess the acuity of the spondylolytic lesion.¹² MRI has been suggested as another imaging modality for evaluation of spondylolysis, but the reported sensitivity ranges from 25% to 86%.^{13–15} In our own series, we found that up to 64% of spondylolysis cases in symptomatic patients can be missed if MRI is the only diagnostic imaging study performed.¹⁶ Consequently, it is standard protocol at our institution to obtain a limited CT scan when presented with a patient whose history and physical examination are suggestive of spondylolysis and an MRI scan that appears “normal.”

Treatment

Most children have resolution of symptoms with conservative treatment consisting of activity modification and bracing. In acute cases, an attempt at achieving bony union is made and activity is restricted. Although reports in the literature are variable, an estimated 75% to 100% of acute unilateral lesions and 50% of acute bilateral lesions heal (Fig. 140.6), whereas essentially no chronic defects heal.⁷ Approximately 90% of athletes return to their prior level of sports participation at an average of 5 to 6 months,¹⁷ which suggests that in many cases a fibrous union is sufficient for symptomatic improvement because the rates of returning to sport exceed the rate of achieving bony union.⁷

No consensus has been reached regarding the type of bracing or protocol for wear. The Boston thoracolumbar sacral orthosis, an antilordotic brace, and braces with a thigh cuff have all been

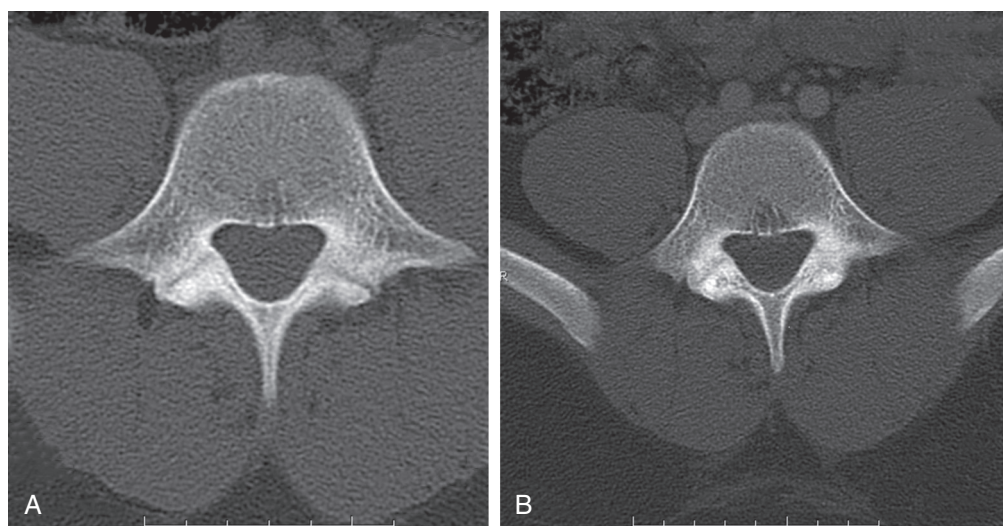


Fig. 140.6 (A) The presence of a bilateral pars defect on a noncontrast computed tomography (CT) scan consistent with a diagnosis of acute spondylolysis. (B) A subsequent CT scan with contrast-confirmed healing of the defect after successful conservative treatment. (Courtesy Children’s Orthopedic Center, Los Angeles, CA.)

used. Some authors suggest that braces be worn for 23 hours a day for 4 to 6 months.¹⁸ In general, when we are attempting to heal the fracture, our preferred treatment of patients with acute spondylolysis is to begin with an antilordotic brace worn any time the patient is weight bearing, including a thigh cuff if tolerated, for the first month. A lumbosacral orthosis without a thigh cuff provides very little immobilization to L5-S1. In acute cases, bracing is continued for approximately 3 months, and a limited CT scan is then performed to evaluate for bony healing prior to return to sport.

In chronic cases in which the bone does not heal, conservative treatment is based on symptoms. A variety of braces prevent excessive hyperextension of the lumbar spine. In some cases, patients can return to their prior level of activity while using such a brace. Core strengthening may play a role in long-term prevention of symptoms.

In cases in which the patient has unresolved pain after more than 6 months of conservative treatment, intolerance of conservative treatment, a greater than 50% slip of the spondylolisthesis, or neurologic symptoms, surgical intervention is considered. Numerous techniques have been described both for direct repair and for fusion of the involved segment. A direct repair should not be considered in cases of spondylolisthesis. Techniques for direct repair include the Buck technique of direct screw fixation across the pars defect, wiring between the transverse and spinous process, and pedicle screws with an attached sublaminar hook described by Kakiuchi.¹⁹⁻²¹ Prior to consideration of a direct repair, an MRI scan should be obtained to evaluate the integrity of the intervertebral disk. If the intervertebral disk is abnormal, a fusion is preferred. Fusion may consist of a posterior only approach, with or without a transforaminal lumbar interbody fusion. Another option for fusion is an anterior and posterior fusion for additional stability. A great deal of controversy continues to exist with regard to in situ fusion versus reduction of the spondylolisthesis. The increased neurologic risks of reduction must be weighed against the risks of implant failure, slip progression, and pseudarthrosis associated with in situ fusion.^{22,23} See Fig. 140.7 for our treatment algorithm for spondylolysis/spondylolisthesis.

Results

Rates of return to sports participation after surgical treatment are high, ranging from 80% to 100%.²⁴ Nevertheless, it is an area of high stress, and implant complications are frequently observed, with posterior fusion techniques resulting in a high reoperation rate, up to 47% in one series, with many of these presenting several years after the initial procedure.²⁵ This may reflect that the population of patients pursuing operative management often represents a subset of very competitive athletes who return to a high level of activity. For this reason, in many of these cases, our center and others have begun favoring the more aggressive approach of a combined anterior and posterior fusion. Fig. 140.8A and B shows an example of a patient treated with fusion who returned to a high level of sports participation.

APOPHYSEAL RING FRACTURES

The vertebral ring apophysis appears at the age of 6 years and fuses to the vertebral body at approximately the age of 17 years.²⁶

Mechanical stress on the apophyseal ring can lead to a fracture of the vertebral growth plate, resulting in an apophyseal ring fracture. This clinical entity, unique to the immature pediatric spine, causes pain similar to that of a disk herniation.

History and Physical Examination

Pain is usually described as acute in onset and is often severe. Radicular pain is less common than in adult disk herniations because the injury is less likely to have a lateral component. A straight-leg raise test may be positive.

Imaging

This injury is often difficult to appreciate on plain radiographs, although such radiographs are generally obtained as an initial step in evaluating the patient with back pain. A CT scan allows accurate characterization of the osseous injury (Fig. 140.9). MRI is generally obtained as well in these cases to allow for evaluation of the adjacent intervertebral disk and nerve roots. However, MRI scans frequently miss this injury or are misread as a disk herniation because the thin bony component of the apophysis may not be appreciated (see Fig. 140.9).

Treatment

In contrast to herniated disk material, the posterior endplate lesion is typically not absorbed. Although injuries with very small osseous fragments may be treated conservatively, such treatment usually results in continued pain. The mainstay of treatment is surgical excision of the protruding fragment. A recent report by Higashino et al.²⁷ supports favorable long-term outcomes in patients with these injuries.

CHANCE FRACTURES

Chance fractures consist of a flexion-distraction injury with failure of both the anterior elements in flexion and the posterior elements in distraction. They were described by Chance as a traumatic horizontal splitting of the spine.²⁸ Chance fractures are often unstable and require surgical management. Consequently, differentiating between a stable compression fracture and a three-column Chance fracture is imperative for proper management of these injuries. Our institution has reported on several patients whom we cared for with unstable Chance fractures that were initially misdiagnosed as compression fractures, with a mean delay in treatment of 3 months.²⁹

History and Physical Examination

Sports injuries are one of the most common mechanisms of injury for pediatric Chance fractures, alongside motor vehicle accidents and falls. These injuries are accompanied by the acute onset of pain and a high incidence of neurologic injury. Arkader et al. reported a 43% incidence of neurologic injury in a multicenter series of pediatric patients with Chance fractures. Of those with neurologic deficits, only 53% made a full recovery.³⁰ Consequently, a thorough neurologic exam is imperative. In addition, as these injuries may be misdiagnosed as compression fractures, it is important to have a high index of suspicion for possible Chance fractures when evaluating patients with a known

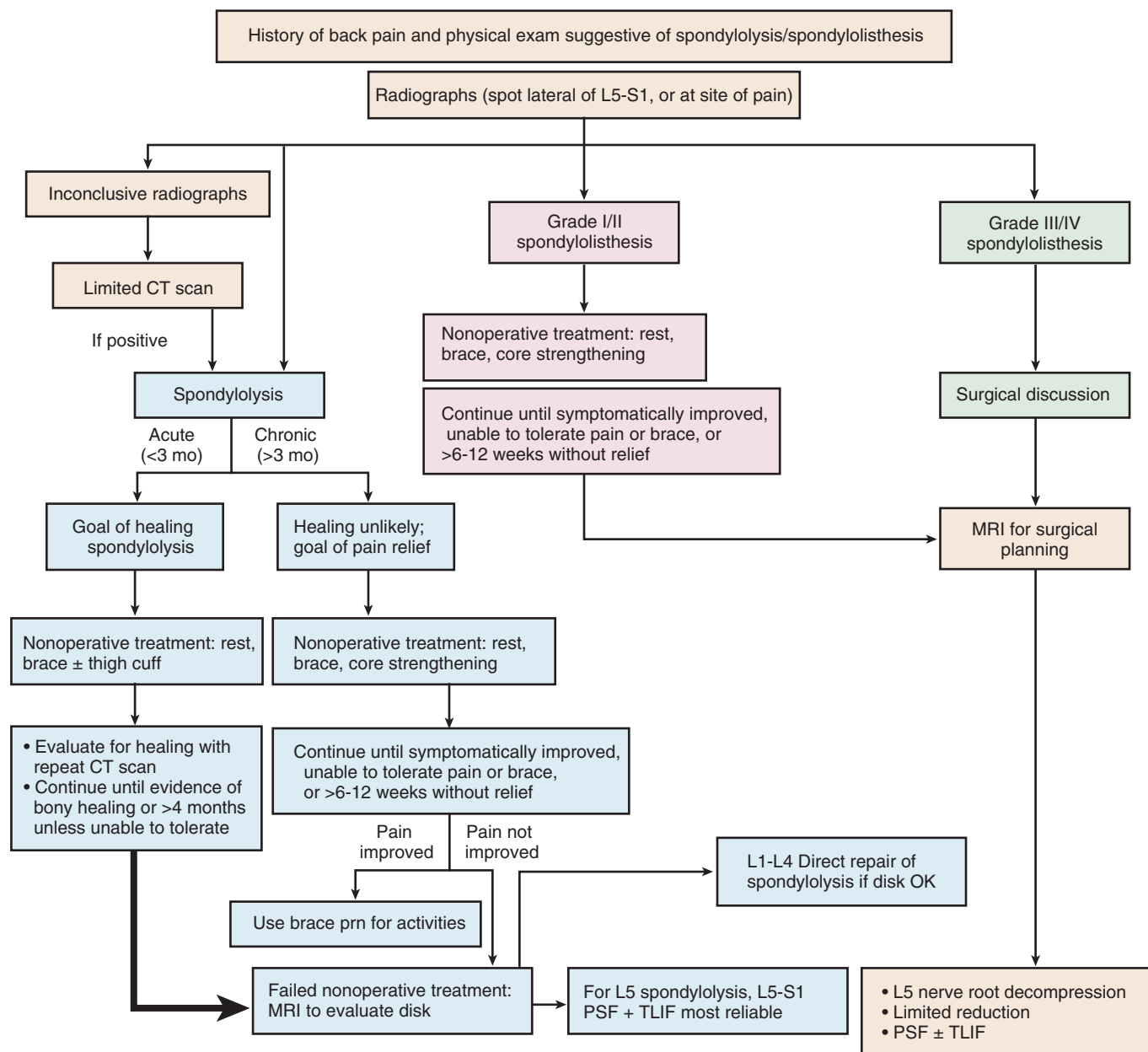


Fig. 140.7 A treatment algorithm for spondylolysis/spondylolisthesis. *CT*, Computed tomography; *MRI*, magnetic resonance imaging; *prn*, as needed; *PSF*, posterior spinal fusion; *TLIF*, transforaminal lumbar interbody fusion.

flexion injury of the anterior column. Clinical concern should be raised if there is tenderness to palpation posteriorly or any swelling or step-off to suggest involvement of the posterior elements with a three-column injury. In addition, there may be a kyphotic deformity at the level of the injury. This may be present initially, depending on the magnitude of the injury, or may develop in cases that are misdiagnosed in a more delayed fashion.

Imaging

The initial evaluation of spine trauma through the thoracolumbar region typically consists of an anteroposterior (AP) and lateral radiograph of the thoracolumbar spine. In cases where a neurologic deficit is present, an MRI of the spine is also obtained. This will reveal the extent of the ligamentous injury. A CT scan is often helpful to define the extent of the osseous injury and

differentiate a purely ligamentous Chance fracture from one with a significant osseous component.

Treatment

Arkader et al. showed improved outcomes in operatively managed pediatric Chance fractures when compared with those treated nonoperatively. In their series, 45% of patients had a good outcome (defined as no chronic pain or neurologic deficit) in the nonoperative group compared with 84% in the operative group.³⁰ Nevertheless, in a neurologically intact patient where there is an osseous rather than a ligamentous injury posteriorly, successful treatment may be possible with a thoracolumbar brace that is fitted so that it provides support at the level of injury to prevent development of kyphosis. In operatively managed patients, treatment typically consists of a level 1 or 2 fusion at the level



Fig. 140.8 (A and B) This 17-year-old female with an L5 spondylolysis was in severe pain despite months of conservative management. Following a successful anterior and posterior spinal fusion of L5-S1, she returned to competitive dance. Clinical photos demonstrate that she maintained excellent flexibility and range of motion following this one-level fusion.

of injury, though more levels may need to be included in cases where significant kyphotic deformity is present.

SACRAL FACET FRACTURES

Sacral facet fractures are a rare injury but an important diagnosis to recognize. They should be considered in the young athlete who has localized back pain with extension. This diagnosis may be easily missed on plain radiographs, bone scans, and MRI, which are frequently negative in these cases.^{31,32} CT is the imaging modality of choice in cases in which this diagnosis is suspected. An example is shown in Fig. 140.10. Authors of a recent case series of elite athletes reported on treatment with removal of the intra-articular fracture fragments by a minimally invasive muscle-sparing approach. In cases where the fragment is removed early, resolution of pain and return to full sports is anticipated, while if long-standing there may be degenerative changes to the facet joint that develop leading to a more negative outcome.³²

SCOLIOSIS

Scoliosis is defined as curvature of the spine of greater than 10 degrees. Although most cases of scoliosis are idiopathic, idiopathic scoliosis is a diagnosis of exclusion. The clinician should be

careful to consider other diagnosis such as neurofibromatosis, Marfan syndrome, or Ehlers-Danlos syndrome, with which the scoliosis may be associated. Many patients with adolescent idiopathic scoliosis—which has a prevalence of nearly 3%—participate in sports.³³ Once other potential concomitant conditions have been eliminated, idiopathic scoliosis is not a contraindication to participation in sports.

History and Physical Examination

At presentation, approximately 20% of patients with scoliosis have back pain, which is similar to the percentage of adolescents who have back pain in general.³⁴ For back pain that is localized, constant, progressive, predominantly occurs at night, or limits activities, further evaluation with MRI is warranted. The clinician should also ask about constitutional symptoms and any bowel or bladder issues. These atypical symptoms necessitate further evaluation for other causes and for spinal cord compression.

Physical examination can be divided into three components: (1) assessment of the curve, (2) evaluation for any neurologic concerns, and (3) examination for any stigmata that suggest a cause that is not idiopathic. The Adams forward bend test then allows assessment of the rotation. Prominence should be noted on the same side as the convexity of the curve. In cases in which a lumbar prominence is noted on the concavity, a leg length

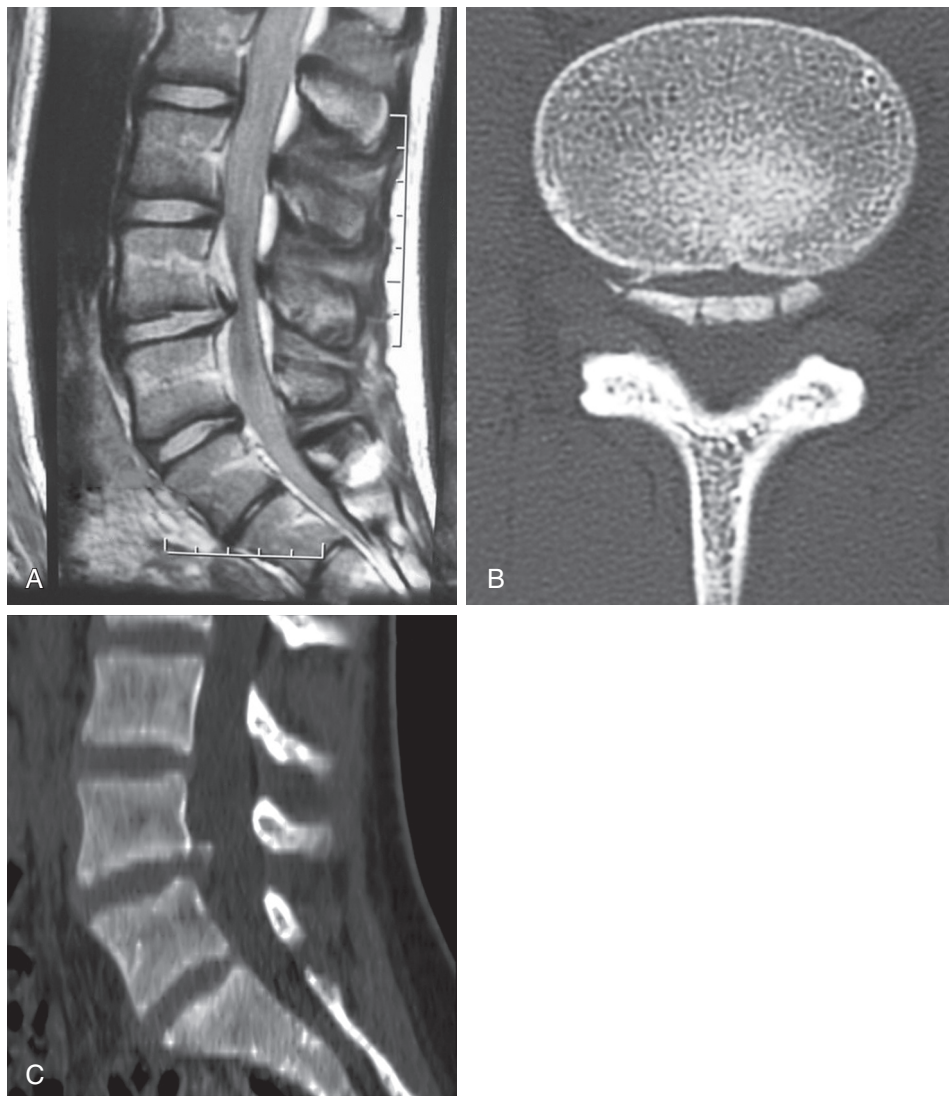


Fig. 140.9 Magnetic resonance imaging (A) gives the appearance of a disk herniation, but a computed tomography scan (B and C) clearly shows that it is in fact an apophyseal ring fracture. (Courtesy Children's Orthopedic Center, Los Angeles, CA.)

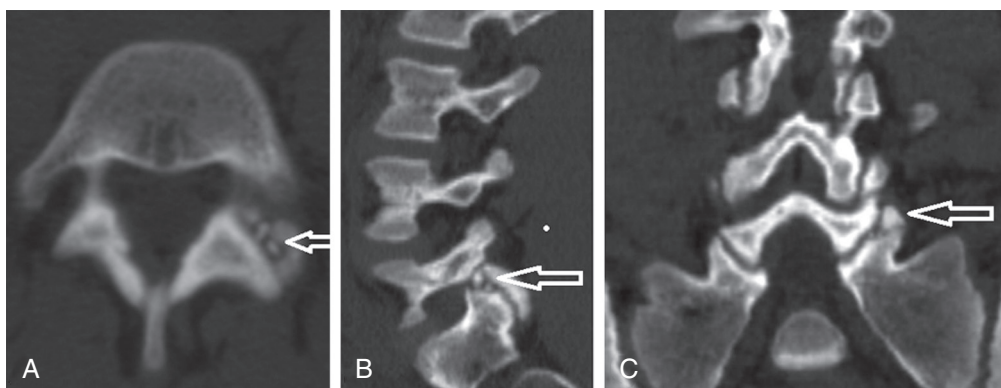


Fig. 140.10 Magnetic resonance imaging scan (A) and A computed tomography scans (B and C) depicting a sacral facet fracture (arrows). (Courtesy Children's Orthopedic Center, Los Angeles, CA.)

discrepancy should be suspected, and the curve may be compensatory. With the patient bent forward, scoliometer readings can be obtained. Although highly variable, a reading of 7 degrees on a scoliometer is considered an indication to obtain a radiograph and correlates to an approximately 20-degree curve.³⁵

After assessment of the curve, a focused neurologic evaluation should be performed (see Box 140.2). Any change in sensation or strength, gait abnormality, foot deformity, unequal reflexes, or sustained clonus suggests a cause that is not idiopathic. Cutaneous findings of axillary freckles, more than three café-au-lait spots, or neurofibromas suggest neurofibromatosis. A high-arched palate, ligamentous laxity, and pectus deformities warrant consideration of Marfan syndrome.

Imaging

If the clinical examination suggests the possibility of scoliosis, then initial radiographic evaluation consists of a standing posteroanterior and lateral radiograph of the thoracolumbar spine. This radiograph ideally includes the pelvis and is not collimated to allow for evaluation of closure of the triradiate cartilage and the Risser sign, which are important components in assessing the growth remaining and the risk of progression. Care should be taken to evaluate for concomitant pathologic conditions on the lateral radiograph. In particular, in persons with symptomatic spondylolysis/spondylolisthesis, an associated scoliosis is present in 25% to 40% of cases, but it may be easily overlooked if the focus is entirely on the spinal curvature.³⁶

Attention should also be directed to the sagittal contour of the patient with idiopathic scoliosis. Due to the three-dimensional nature of scoliosis, rotation and resultant hypokyphosis are generally observed through the thoracic region. If this area is instead kyphotic, one should suspect other etiologies. In cases in which a cause other than idiopathic scoliosis is suspected, an MRI should be obtained. Although recommendations vary, indications for an MRI include (1) neurologic abnormality (weakness/abnormal reflexes), (2) severe pain, (3) young patients (younger than 11 years) with curves greater than 20 degrees, (4) atypical patterns (i.e., a left-sided thoracic curve, congenital scoliosis, short angular curves, or a severe deformity >70 degrees), and (5) rapid progression (i.e., >1 degree per month).³⁷

In cases where there is a question as to the amount of growth remaining, a radiograph of the hand may be obtained to better assess skeletal maturity. Evaluation of this radiograph with a modified classification described by Sanders et al. has been found to be a more accurate predictor of risk of progression than Risser staging or Greulich and Pyle bone age.³⁸

Treatment

Treatment is based on the risk of curve progression. The two major components in the evaluation are curve magnitude and skeletal maturity (i.e., the amount of growth remaining). Lonstein and Carlson have reported that if the condition is not treated, 68% of patients who are at Risser level 0 to 1 with a 20- to 29-degree curve will progress, whereas only 23% of patients who are at Risser level 2 to 4 with a curve of the same magnitude will show signs of progression.³⁹ Currently none of the genetic testing available reliably allows identification of the

curves that will progress, as opposed to curves that will not progress.

Nonoperative treatment consists predominantly of bracing. Protocols and indications vary with much of the debate centering around the contradictory evidence with regard to bracing, which has been complicated by compliance issues. In 2010, Katz et al.⁴⁰ reported on 82% of patients who wore a brace more than 12 hours a day did not have a progression of their curve, compared with 31% who did not progress with brace wear of fewer than 7 hours per day.⁴⁰ Another landmark article was published in 2013 in the *New England Journal of Medicine* by Dolan and Weinstein reporting the results of the BRAIST trial.⁴¹ In this multicenter, randomized prospective study between observation and bracing, the success rate was 72% in the braced group compared with 48% in the observation group.⁴¹ In general, bracing with a thoracolumbosacral orthosis (Boston or Rigo Cheneau brace) can be considered for curves of 25 to 40 degrees, with substantial growth remaining (Risser level 3 or less; though more accurately assessed by Sanders staging) and a curve apex below T7.

For curves that reach 50 degrees, surgical intervention is generally recommended. This recommendation is based on the high rate of progression of curves with that magnitude even after skeletal maturity and the ultimately significant cardiopulmonary effects.³³ Currently posterior spinal fusion with pedicle screw fixation is the most common surgical treatment for idiopathic scoliosis. Complications include ileus (6%), early infection (1% to 2%), late infection (up to 5%), pseudarthrosis (3%), and neurologic deficit (0.4% to 0.7%).³⁷

Scoliosis and Sports Participation

Numerous studies have demonstrated that exercise is not sufficient to inhibit the progression of idiopathic scoliosis. Although it may not alter the natural history, swimming has been shown to help maintain flexibility of the spine, as well as strength and endurance.⁴² Although it is widely recognized that scoliosis should not present a contraindication to sports participation, many questions remain about when it is appropriate for patients to return to sports in cases in which athletes have undergone surgical treatment. Survey responses from 261 pediatric spine surgeons indicate that the majority of these surgeons allow children to return to noncontact sports participation between 6 months and 1 year after spinal fusion for adolescent idiopathic scoliosis.⁴³ Even more variable was the timing for return to contact sports. In most cases of uncomplicated posterior spinal fusion for idiopathic scoliosis, we allow return to sports as tolerated, including contact sports, at around 3 months after the operation. However, this return is dependent on the understanding of both the patient and parent that some very small increase in the risk of a catastrophic spinal injury may exist, along with the agreement to undertake this shared risk.

Varying reports have been made regarding the long-term impact of surgical treatment for scoliosis on sports participation. In 2002, a survey of patients with scoliosis who were treated both operatively and nonoperatively reported no significant difference in sports participation, but both groups had decreased participation compared with control subjects.⁴⁴ More recently, Fabricant et al.⁴⁵ reported a correlation between the distal level

of the fusion and return to sports after posterior spinal fusion. In their study, only 59% returned to their preoperative level of athletic activity at an average of 5.5 years of follow-up.⁴⁵ Their study did not include a nonoperatively treated group for comparison. Selective posterior spinal fusions, more stable constructs with pedicle screws, and postoperative protocols with earlier return to physical activity may help improve outcomes, but as of yet, data on this topic are limited.

ATLANTOAXIAL INSTABILITY

Atlantoaxial instability may be either acute, as a result of a traumatic event, or chronic. Acute atlantoaxial instability may occur as a result of severe flexion of the cervical spine. An ADI of 5 mm or more in children in the setting of an acute injury may signify that the transverse ligament has been injured and that instability is present. Magnetic resonance imaging (MRI) can further elucidate the extent of the injury. If the transverse ligament is injured, posterior spinal fusion of C1 and C2 is usually necessary to prevent further compromise to the spinal cord.

ATLANTOAXIAL INSTABILITY IN DOWN SYNDROME

More commonly the physician is faced with management of chronic atlantoaxial instability in patients with Down syndrome. This instability may be either symptomatic or asymptomatic. The estimated incidence of asymptomatic atlantoaxial instability in persons with Down syndrome is up to 22%, whereas only 2% to 3% of persons with Down syndrome have symptomatic atlantoaxial instability.⁴⁶ The literature currently does not include any reports of children with Down syndrome without preceding neurologic symptoms who have had a traumatic injury with sports participation that resulted in progression to symptomatic instability. The history and physical examination are especially important in the evaluation of these patients, as even patients with normal cervical spine radiographs may develop instability over time.⁴⁷

History and Physical Examination

The history is key in this patient population because the physical examination may be challenging. The revised American Academy of Pediatrics 2011 guidelines mark a shift away from routine radiographic screening and emphasize the need for a physical exam and parental counseling to monitor for development of any symptoms on a biannual basis.⁴⁸ Symptoms to inquire about include neck pain, fatigability, difficulty walking, abnormal gait, and worsening coordination/clumsiness or change in bowel or bladder habits. Physical examination should include evaluation of neck range of motion (ROM), gait, strength, spasticity, hyperreflexia, and clonus.

Imaging

Both the screening and management of atlantoaxial instability in persons with Down syndrome have become extremely controversial. The American Academy of Pediatrics has retracted its recommendation for assessment of potential atlantoaxial

instability and currently advises against imaging in asymptomatic persons.⁴⁸ In 2014, the Special Olympics also modified their requirements and no longer require routine radiographs for athletes. Instead they now rely on the completion of a medical form that includes a review of symptoms for adverse neurologic effects, including those that could result from spinal cord compression or symptomatic atlantoaxial instability. These forms are to be completed a minimum of every 3 years, though many programs require them on a more frequent basis. Radiographs should be obtained if there are any symptoms suspicious for this. The guidelines established by Special Olympics state that an ADI of greater than 4.5 mm is considered to be atlantoaxial instability.⁴⁹ Other authors have suggested that measurement of the space available for the cord (spinal canal width) of less than 14 mm is a more relevant predictor of neurologic injury, though this remains controversial in younger age groups. Nakamura et al. reported on use of the C1 to C4 ratio of space available for the cord (SAC), which had a high inter and intrarater reliability and can account for change with age and growth.⁵⁰ They reviewed more than 250 children with Down syndrome and found that a C1/C4 SAC of less than 0.8 was associated with a high chance of developing myelopathy.⁵⁰ If either an abnormal space available for the cord or an abnormal ADI is detected, further evaluation by a pediatric spine surgeon is warranted. In cases in which the ADI is greater than 10 mm or the child is symptomatic, an MRI scan may be helpful in the evaluation.

A similar issue for pediatric athletes with Down syndrome is the consideration of atlantooccipital hypermobility. Atlantooccipital instability has been underappreciated and should be considered when evaluating radiographs in this population.⁵¹ A Power ratio of greater than 1.0 signifies anterior instability.⁵² In addition, posterior instability has also been described, but the clinical significance of this instability is as yet unclear.⁵³

Decision-Making Principles

Activity restriction on the basis of asymptomatic atlantoaxial instability is a topic of great controversy. Opponents argue that no cases have been reported of patients with isolated radiographic atlantoaxial instability who sustained a neurologic injury as a result of sports participation.⁵⁴ In addition, no cases have been reported of asymptomatic atlantoaxial instability progressing to symptomatic atlantoaxial instability as a result of a sports-related injury.⁵⁵ Based on their own internal review, the Special Olympics committee has consequently transitioned from routine screening radiographs to a medical form that emphasizes a history and review of symptoms to check for any evidence of spinal cord compression that could be due to atlantoaxial instability. If any of those symptoms are present, a flexion extension lateral cervical spine radiograph is obtained. For patients with an ADI of greater than 10 mm, spinal fusion is generally recommended. Fig. 140.11 shows our preferred treatment in the screening and management of asymptomatic persons with Down syndrome.

Currently no recommendations exist regarding atlantooccipital hypermobility in terms of clearance for the Special Olympics or other sporting activities in the young athlete with Down syndrome, but it should be considered when monitoring for the more common atlantoaxial instability.⁵¹

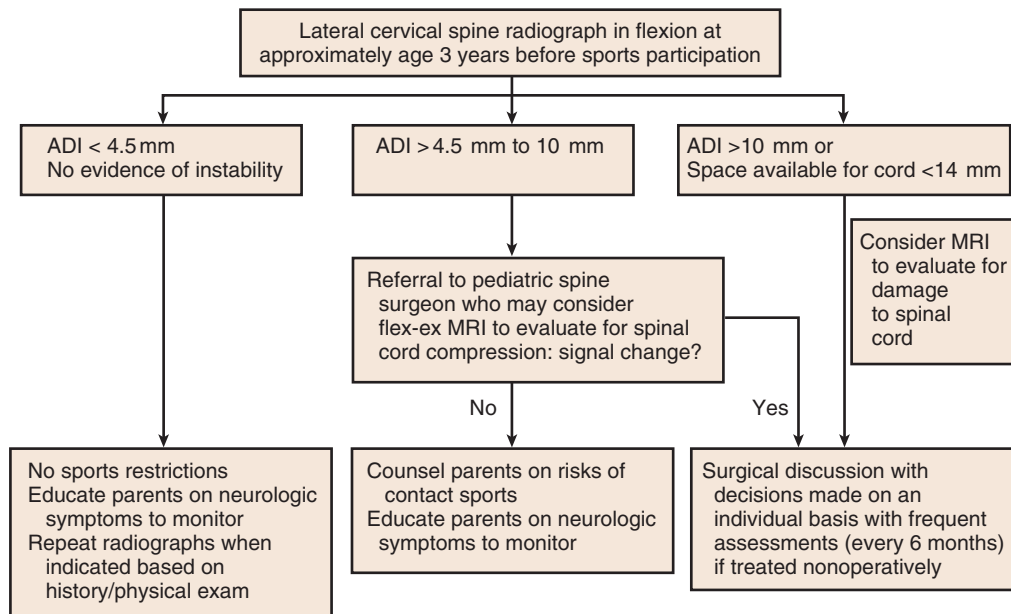


Fig. 140.11 The authors' screening protocol for asymptomatic patients with Down syndrome. ADI, Atlanto-dens interval; flex-ex, flexion-extension; MRI, magnetic resonance imaging.

Treatment

Aside from activity restrictions, no other nonoperative treatments are available for atlantoaxial instability. Operative management consists of posterior arthrodesis. Prior to fusion, particular attention needs to be given to the occipitocervical articulation. If there is evidence of instability observed at this level, in cases where the C1/C2 alignment is irreducible, or where decompression is needed of that region is required, the occiput is generally included in the fusion. Many techniques for atlantoaxial fusion have been described, including the use of wires, plates, and transarticular screws with either a rib or iliac crest graft. For smaller children, two commonly used techniques are the Mah modified Gallie technique and the Brooks technique. In the Mah modified Gallie technique, a wire is passed under the posterior arch of C1, looped over a threaded Kirschner wire that is placed in the spinous process of C2, and then tightened to secure a rectangle of corticocancellous autograft in position.⁵⁶ In the Brooks technique, two sublaminar wires are passed under both C1 and C2 and used to secure two trapezoids of the iliac crest graft that are wedged between C1 and C2.⁵⁷ Halo vest immobilization is required until bony fusion has been achieved. Screw fixation has also been utilized, particularly in older children. In cases where the occiput needs to be included, the authors preference is fixation with a plate. An important distinction is the cases of instability that are reducible, which can be positioned in extension to restore normal anatomic alignment, thus providing adequate space for the cord, and those that are not reduced by positioning in extension. In the latter cases, a concomitant decompression is often needed (Fig. 140.12A–C).

Results

Fusion rates in the setting of atlantoaxial instability in persons with Down syndrome have historically been poor and range

from 40% to 80%.⁵⁸ An improvement in these results has been reported by Menezes and Ryken⁵⁹ by incorporating the occiput when indicated and implementing prolonged immobilization with a halo postoperatively, with recommendations of 4 months for C1–C2 fusion and 6 months when the occiput is included. Complication rates are high for this procedure. Segal et al.⁶⁰ have reported a 100% complication rate and an 18% mortality rate.

ROTATORY ATLANTOAXIAL SUBLUXATION

A common cause of acute torticollis is rotatory atlantoaxial subluxation, which occurs when the lateral mass of the atlas locks behind the ipsilateral lateral mass of the axis. It is most often traumatic in nature in healthy children, although other possible causes exist, such as Grisel or Sandifer syndrome. Rotatory atlantoaxial subluxation may occur after even minor trauma during sports participation or even with coughing episodes. It also may be associated with conditions that feature ligamentous laxity in this region, including Down syndrome, Marfan syndrome, or rheumatoid arthritis.

History and Physical Examination

On presentation, patients have a “cock robin” torticollis with their head tilted toward one shoulder and their chin rotated in the opposite direction. Patients may report neck pain and occasionally occipital neuralgia or symptoms of vertebrobasilar insufficiency. This condition is differentiated from muscular torticollis both by the acuity and by the absence of a palpable sternocleidomastoid on the shortened side. Conversely, the lengthened sternocleidomastoid may feel tight from being stretched. Most affected children are neurologically intact, but a thorough neurologic examination should be performed.

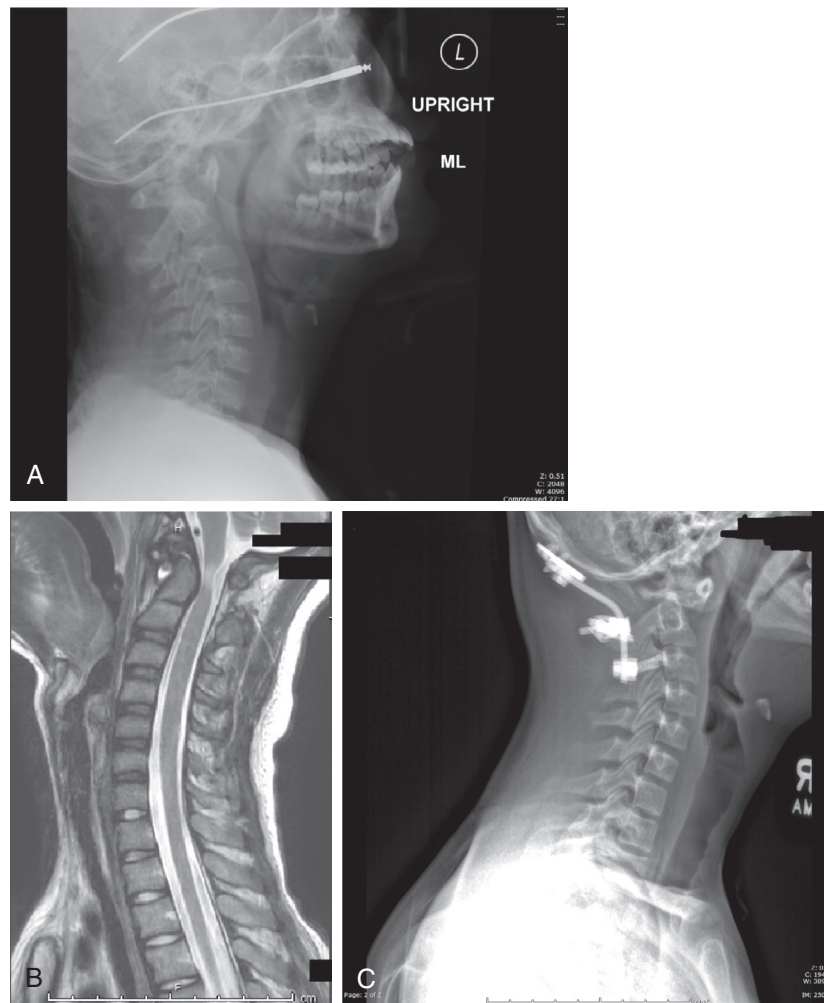


Fig. 140.12 (A-C) A 14-year-old male with Down syndrome who presented after several months of worsening gait instability and weakness. (A) Lateral radiograph showed marked anterior displacement of C1 on C2 which did not reduce. (B) Magnetic resonance imaging demonstrated compression and cord signal change at the level of the C1. The patient underwent decompression and occipitocervical fusion, after which he made a full neurologic recovery.

Imaging

Radiographic evaluation should include an open-mouth AP and lateral radiograph of the cervical spine. The technician should take the image as a true lateral view of the skull, which results in images that are more easily interpreted and show a true lateral view of the axis and rotation of the remainder of the spine. For further evaluation or if the diagnosis is unclear, a CT scan can provide additional clarity (Fig. 140.13). Dynamic CT scans allow for a more complete assessment of this area and avoid an ambiguous test result due to positioning, but should be limited to the upper cervical spine (occiput to C3) to minimize radiation.

Treatment

Treatment is time dependent because the subluxation becomes more difficult to reduce with the passage of time. In addition, persistent subluxation may damage the transverse ligament and result in subsequent instability or recurrence if a reduction is obtained.⁶¹ In general, if the diagnosis is made and treatment starts 1 month or less from the time of the injury, then nonoperative management with either a cervical collar or head halter traction

is often sufficient. In patients in whom treatment is initiated less than 1 week after the onset of symptoms, immobilization in a soft collar for a few days is often sufficient to both obtain and maintain a reduction. For patients who begin treatment 1 to 4 weeks after the onset of symptoms, head halter traction can be used. If reduction is attempted with head halter traction, the position can be maintained with the use of a noninvasive halo. If these measures fail or if the injury occurred more than 4 weeks prior to the beginning of treatment, closed reduction with a noninvasive halo may be attempted, with reduction likely once the head is rotated past midline. A CT scan confirms the reduction. Use of the noninvasive halo may be continued for 4 to 6 weeks to allow scarring and maintenance of the reduction.⁶² If instability or loss of reduction occurs, then repeat reduction with the noninvasive halo followed by surgical stabilization with posterior atlantoaxial fusion is indicated. Although it was once thought that late presentation of rotatory atlantoaxial instability frequently resulted in irreducible or recurrent subluxation and fusion was frequently needed, two recent series have challenged this assertion. Glotzbecker et al. reported a 73% success rate in patients treated nonoperatively with a mean delay of presentation

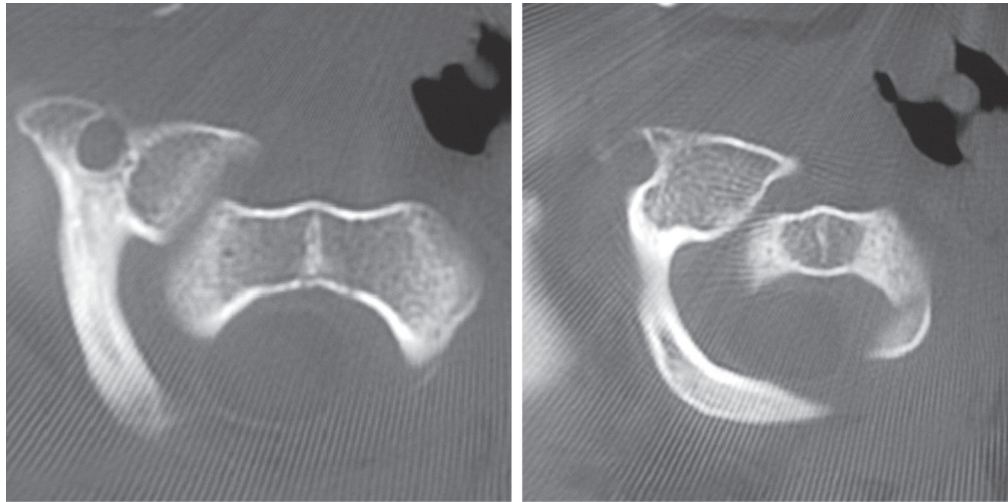


Fig. 140.13 A computed tomography scan showing that C1 is subluxated on C2 with the lateral mass of the atlas locked behind the ipsilateral lateral mass of the axis. (Courtesy Children's Orthopedic Center, Los Angeles, CA.)

of 18 weeks.⁶³ Chechik et al. reported a 100% success rate with patients treated nonoperatively with a mean delay of 11 weeks to presentation.⁶⁴ Although the authors recommend that this condition be treated promptly to optimize results, it appears that even in cases of delayed presentation, a trial of conservative therapy is warranted.

ODONTOID FRACTURES

Odontoid fractures account for 10% of all cervical spine injuries in children and the majority of pediatric cervical spine fractures.⁶⁵ In children younger than 7 years, the fracture occurs through the synchondrosis at the base of the dens. This position can make the diagnosis more challenging because the synchondrosis may be mistaken for a fracture or, conversely, the fracture may be assumed to be the synchondrosis and overlooked (see Fig. 140.14).

Classification

Odontoid fractures in children are described with use of the adult classification system of Anderson and D'Alonzo. Type I is an avulsion fracture of the tip of the dens. Type II is the most common and is through the base of the dens. In type III, the fracture extends into the vertebral body.

History and Physical Examination

Odontoid fractures occur more commonly from flexion mechanisms but have also been described from forced hyperextension. Although most authors have reported that pediatric odontoid fractures are not associated with neurologic injury, one series reported neurologic injury in 8 of 15 children younger than 6 years who had odontoid fractures.⁶⁶ A thorough physical examination should be performed to evaluate for this possibility. Typically these children have neck pain and decreased ROM. Direct point tenderness is not necessarily present.

Imaging

The synchondrosis at the base of the dens should be fused by age 6 years. After this age, any evidence of a radiographic abnormality

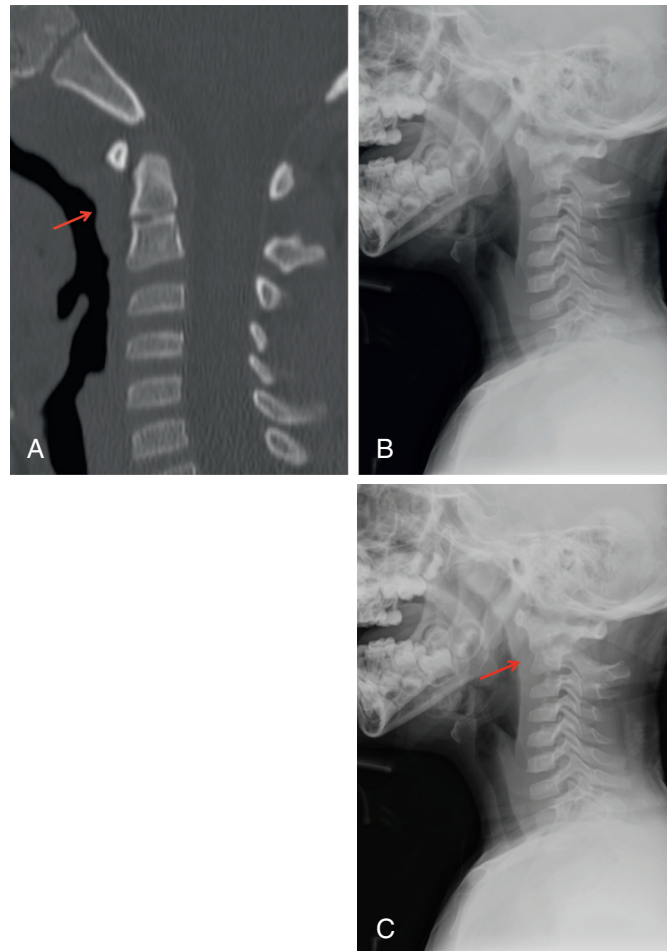


Fig. 140.14 A 4-year-old male who had sustained a fall on his head had an initial computed tomography scan (A) and radiograph (B) that were read as negative. The odontoid fracture (arrows) was missed because it was assumed to be just his normal physis despite the associated swelling. He presented 2 weeks later to his primary care physician with neck pain; a lateral radiograph taken at that time (C) shows significant displacement through the fracture site. Treatment consisted of closed reduction and halo placement. (Courtesy Children's Orthopedic Center, Los Angeles, CA.)

in this region suggests a fracture. In children younger than 7 years, great care must be taken to not mistake normal anatomy for an acute injury or conversely to neglect a fracture by attributing it to normal development. Assessment for swelling in the retropharyngeal space can be helpful in this differentiation. Nevertheless, the clinician should keep in mind that if the patient is crying, the child will have an apparent increase in the width of the retropharyngeal space. In situations in which the diagnosis is unclear and clinical suspicion is high, an MRI may be obtained for further evaluation. Fig. 140.14 shows the initial CT scan of a 4-year-old boy who fell on his head and whose odontoid fracture was missed because this fracture was assumed to be just his normal physis despite the associated swelling. He presented to his primary care physician with neck pain 2 weeks later, and a lateral radiograph showed significant displacement through the fracture site.

Treatment

Most of these injuries can be treated by stabilization in a halo vest. The alignment of the fracture is generally improved with the head positioned in slight extension. The amount of acceptable angulation and the potential for remodeling varies with age and remains controversial, although one report of patients younger than 3 years with angulation of greater than 30 degrees showed that remodeling led to a normal structure with no associated sequelae.⁶⁷ In children 7 years of age or older, the fractures behave more like adult fractures and are treated accordingly. Cases of nonunion of type II odontoid fractures in this group have been reported and can be treated with a screw placed via a direct anterior approach.⁶⁸ Consideration should also be given to the possibility of additional ligamentous injury causing instability, which necessitates C1-C2 fusion.

CONGENITAL ANOMALIES OF THE CERVICAL SPINE

Congenital anomalies may involve either failure of formation or segmentation of vertebrae. Numerous anomalies exist, and each case warrants evaluation for concomitant stenosis or instability. Evidence of instability may warrant surgical management and at the very least avoidance of sporting activities. One such clinical entity is os odontoideum. In this condition, the dens is not fused to the body of the C2 vertebrae. Another commonly encountered congenital anomaly is that of Klippel-Feil, the term given to a congenital fusion (or failure of segmentation) of two or more cervical vertebrae.

Classification

Klippel-Feil anomalies are divided into two types. Type I is a multilevel deformity in which a large fusion mass is present, and type II is a fusion of either one or two levels.

History and Physical Examination

Although os odontoideum was once believed to be exclusively a congenital condition, more recently a posttraumatic cause has also been proposed.⁶⁹ Evidence supports both posttraumatic and congenital causes, and in reality, two separate clinical entities may exist.⁷⁰ The clinician may be evaluating a patient because

of an incidental radiographic finding or because pain or neurologic symptoms are present.

Patients with Klippel-Feil syndrome may have the classic triad of a low hairline, a short neck, and decreased cervical ROM, but these findings are present in fewer than 50% of cases.⁷¹ These patients may also have an associated Sprengel deformity due to failure of the scapula to descend. Evaluation of a person with Klippel-Feil syndrome also may be in response to an incidental radiographic finding. In all cases, a thorough evaluation for any neurologic symptoms or physical findings must be performed.

Imaging

At a minimum, imaging consists of AP, lateral, and flexion extension radiographs of the cervical spine. In cases of odontoid abnormalities, improved visualization is obtained with an open-mouth odontoid view. If evidence of instability or concern about instability is present on the basis of neurologic abnormalities, an MRI of the spine is warranted. A flexion-extension MRI is preferred in many cases to evaluate for any subtle changes indicative of dynamic cord compression. Prior to performing the flexion component of the MRI, it is advisable to obtain a neutral MRI and confirm that there isn't already cord signal change that would suggest flexion for the MRI may cause neurologic trauma.

It is also important to differentiate a persistent ossiculum terminale from an os odontoideum. These clinical entities differ in that the os terminale is small in size and located above the transverse atlantal ligament. Consequently, this condition is generally not associated with instability, although there are case reports of this occurring when associated with other syndromes.⁷²

An additional consideration in the patient with congenital anomalies of the spine is the increased incidence of genitourinary issues and congenital heart disease in this patient population. Given the concomitant incidence, it is recommended that these patients should have a cardiac evaluation and renal ultrasound prior to consideration of sports participation. For example, the presence of renal abnormalities, such as a solitary kidney, may pose additional risks with sports participation.⁷³ Although recommendations vary, a diagnosis and discussion of the risks prior to sports participation are warranted.⁷⁴

Treatment

In cases of os odontoideum in which the patient has neurologic symptoms and evidence of radiographic instability, a cervical fusion of the first and second cervical vertebrae is indicated. In cases of asymptomatic instability, the treatment is controversial, and both radiographic surveillance and surgical management have been recommended. Little evidence supports either decision. In persons with os odontoideum, as well as odontoid agenesis or hypoplasia, participation in contact sports is usually contraindicated. The literature includes multiple reports of injury to patients with previously undiagnosed os odontoideum that resulted in a neurologic injury, including respiratory-dependent quadriplegia.⁷⁵

In persons with Klippel-Feil syndrome, the potential for spinal cord injury is due to the abnormal motion of the adjacent unfused segments. Pizzutillo has reported that more than 90 cases of

neurologic injury have occurred in patients with Klippel-Feil syndrome because of associated occipitocervical anomalies, chronic instability, disk disease, or degenerative joint disease.⁷⁶ More than two-thirds of these cases were associated with a single-level fusion of the upper cervical spine.⁷⁷ Although no consensus has been achieved on this topic, a congenital fusion at level C3 or higher is sometimes considered a relative contraindication to participation in collision sports and warrants evaluation by a pediatric spine surgeon. Conversely, patients with a one-level fusion below the level of C3, a normal neck ROM, the absence of occipitocervical abnormalities, and the absence of instability do not have any contraindication to sports participation.⁷⁵

SPINAL CORD INJURY WITHOUT RADIOGRAPHIC ABNORMALITY

The acronym SCIWORA was coined by Pang and Wilberger⁷⁸ to describe a spinal cord injury with motor, sensory, or combined deficits in the absence of any vertebral fracture or malalignment on plain radiographs or a CT scan. Although SCIWORA has been described in persons of all ages, it is most commonly observed in the pediatric population and accounts for an estimated 19% to 34% of all spinal cord injuries.^{78–80} A great deal of controversy surrounds this condition, including its definition, diagnosis, and optimal management. In many cases, the term is a misnomer because clear evidence of the neurologic injury is observed on MRI. In some series, such as the one reported by Bosch et al.,⁸¹ 50 of the 60 MRI scans obtained were interpreted as normal.³² Conversely, a recently published meta-analysis found that of the 392 cases identified, every patient with a diagnosis of SCIWORA with an MRI scan had evidence of either intraneural or extraneural injury.⁸² In a series by Hamilton and Myles,⁸³ the onset of neurologic symptoms was delayed in 23% of cases (range, 6 to 72 hours).

History and Physical Examination

Patients with SCIWORA have sustained a traumatic event and have a subsequent neurologic deficit. Most cases of SCIWORA involve the cervical spine, but cases involving the thoracic and lumbar spine have also been described.⁸⁴ Motor vehicle accidents, falls, and sports are frequent mechanisms, though a 2012 review of a national database reported sports as the most common, accounting for 40% of the injuries. SCIWORA is most commonly associated with a flexion injury, but distraction and extension injuries have also been described with this diagnosis.⁸⁵ The severity of the neurologic injury is highly variable, ranging from isolated nerve root involvement to complete quadriplegia. The spine is usually tender with the presence of significant local swelling or a hematoma.

Imaging

By definition, the absence of any vertebral fracture or malalignment on plain radiographs or a CT scan must be observed to assign the diagnosis of SCIWORA. MRI scans of the brain and spine are also obtained for persons with this condition to evaluate for evidence of spinal cord injury and any other central nervous system abnormality in the differential. Several anatomic explanations for the more frequent occurrence of SCIWORA in

the pediatric cervical spine have been proposed, including a more horizontal orientation of the facet joints, anterior wedging of the vertebral bodies, and increased elasticity of the ligaments and joint capsules. A sentinel finding on MRI of a ligamentous Chance fracture that may be missed on plain films is significant, as evidenced by a subcutaneous hematoma around the failed posterior elements and/or an increased T2-weighted signal at the interspinous ligament.

Treatment

A wide spectrum of severity is found in this condition, ranging from deficits localized to one nerve root to complete paralysis. Because of the variation in clinical presentations, the appropriate management is difficult to determine. Uses of corticosteroids, mannitol, hyperbaric oxygen, and bracing have all been described.⁸¹ The variable treatment protocols instituted from one institution to the next have further confounded this analysis and made it impossible with the currently available data to ascertain if steroids are of true benefit. Some authors have also argued against bracing or immobilization; they believe it is not indicated based on a lack of associated instability. Other authors, including Launay et al.,⁸² have recommended treatment with immobilization. In their study, they compared a subgroup of 93 patients treated with immobilization only (with no use of steroids or other medications) and divided them into groups who were immobilized for 8 weeks and for 12 weeks. These investigators observed complete recovery in 42% of the patients immobilized for 12 weeks, but in only 34% of the patients immobilized for 8 weeks.⁸²

At our institution, the following protocol is used in the management of persons with SCIWORA: In cases in which evidence of spinal instability exists (i.e., ligamentous injury, including ligamentous Chance fractures), we believe that spinal fusion with instrumentation is the most reliable treatment to protect the patient from further injury and allow early rehabilitation. If no evidence exists of an occult injury that may be associated with instability, then we do not routinely immobilize these patients.

CONCLUSION

Evaluation of the growing spine is associated with unique challenges and concerns. Nevertheless, with appropriate treatment, successful return to sport can often be achieved in pediatric and adolescent athletes.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Jackson RS, Banit DM, Rhyne AL, et al. Upper cervical spine injuries. *J Am Acad Orthop Surg*. 2002;10:271–280.

Level of Evidence:

III

Summary:

The authors describe the unique anatomy of the upper cervical spine, mechanisms of injury, and injury patterns, and review nonoperative and operative treatment of these injuries.

Citation:

Bull MJ. Clinical report: health supervision for children with down syndrome. *Pediatrics*. 2011;128:393–406.

Level of Evidence:

V

Summary:

This article includes guidelines from the American Academy of Pediatrics on caring for patients with Down syndrome and managing associated conditions.

Citation:

Tokunaga S, Ishii Y, Alzawa T, et al. Remodeling capacity of malunited odontoid process fractures in kyphotic angulation in infancy. *Spine*. 2011;36:E1515–E1518.

Level of Evidence:

III

Summary:

The authors performed a retrospective study of odontoid fractures that occurred in children younger than 3 years of age with a minimum of 20-year follow-up; they compared these patients with 127 patients who did not have a history of cervical spine trauma. At final follow-up, patients who had sustained odontoid fractures did not demonstrate a difference in odontoid process tilt angle compared with control subjects.

Citation:

Sankar WN, Wills BP, Dormans JP, et al. Os odontoideum revisited: the case for a multifactorial etiology. *Spine*. 2006;31:979–984.

Level of Evidence:

IV

Summary:

The authors reviewed 519 cervical spine radiographs for cases of os odontoideum. A history of trauma was reported in only 8 of 16 cases, supporting the theory that two separate etiologies may exist for the os odontoideum: posttraumatic and congenital.

Citation:

Grinsell MM, Showalter S, Gordon KA, et al. Single kidney and sports participation: perception versus reality. *Pediatrics*. 2006;118(3):1019–1027.

Level of Evidence:

III/V

Summary:

The authors report a combination of a survey of 135 American Society of Pediatric Nephrology members (level of evidence: V) and a review of literature on sports-related kidney injuries (38 articles; level of evidence: III). Although only one article reported one injury to a single kidney that was managed conservatively without sequelae, 62% of respondents said they would not allow sports participation by patients with a single kidney.

Citation:

Launay F, Leet AI, Sponseller PD. Pediatric spinal cord injury without radiographic abnormality. *Clin Orthop Relat Res*. 2005;433:166–170.

Level of Evidence:

III

Summary:

The authors provide a meta-analysis of 392 published cases of spinal cord injuries without radiographic abnormalities and evaluation of the mechanism of injury, risk factors, treatment, and prognosis.

Citation:

Weinstein SL, Zavala DC, Ponseti IV. Idiopathic scoliosis: long-term follow-up and prognosis in untreated patients. *J Bone Joint Surg Am*. 1981;63A:702–712.

Level of Evidence:

III

Summary:

The authors provide a classic article on the prognosis of persons with idiopathic scoliosis that is untreated with an average follow-up of 39 years.

Citation:

Fabricant PD, Admoni SH, Green DW, et al. Return to athletic activity after posterior spinal fusion for adolescent idiopathic scoliosis: analysis of independent predictors. *J Pediatr Orthop*. 2012;32(3):259–265.

Level of Evidence:

II

Summary:

The authors performed a retrospective cohort study of patients with adolescent idiopathic scoliosis in which 60% returned to sports at an equal or higher level of physical activity after surgery. A distal level of fusion, Lenke classification, and postoperative SRS-22 scores were each independent predictors of rate of return to the preoperative level of activity.

Citation:

Hu S, Tribus CB, Diab M, et al. Spondylolisthesis and spondylolysis. *J Bone Joint Surg Am*. 2008;90A:656–671.

Level of Evidence:

Instructional course lecture

Summary:

The authors provide a review on nonoperative and operative management, and prognosis of patients with spondylolisthesis and spondylolysis.

Citation:

Higashino K, Sairyo K, Katoh S, et al. Long term outcomes of lumbar posterior apophyseal end plate lesions in children and adolescents. *J Bone Joint Surg Am*. 2012;94A:e74.

Level of Evidence:

III

Summary:

This study demonstrated a favorable outcome for pediatric patients with posterior endplate lesions in both operatively and nonoperatively managed cases with a mean follow-up of 14 years.

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